

1. INTRODUCTION

The **Shop Easy – E-Commerce Order Management Database System** is designed to manage the daily operations of an online shopping platform through a centralized database. The system stores and manages customer information, product details, inventory, suppliers, orders, payments, shipments, and customer reviews. By replacing manual record management with a structured database solution, the organization can improve operational efficiency, maintain accurate records, reduce data redundancy, and generate meaningful business reports.

2. Business Scenario Analysis

A rapidly growing e-commerce company requires a reliable database system to support its expanding business operations. The organization needs a centralized solution capable of handling customer registration, product management, inventory tracking, order processing, payment management, shipment tracking, and customer reviews.

Currently, managing these operations manually or using separate files leads to data duplication, inconsistent records, and delays in processing customer requests. To overcome these challenges, the company plans to implement the **Shop Easy – E-Commerce Order Management Database System**, which stores all business information in a single database and ensures efficient management of daily activities.

3. Business Problem

The organization faces several operational challenges due to the lack of a centralized database system.

The major problems include:

- Manual management of customer information.
- Duplicate and inconsistent data.
- Difficulty maintaining inventory records.
- Delays in processing customer orders.
- Inefficient payment tracking.
- Poor shipment monitoring.
- Difficulty generating business reports.
- Increased chances of human errors.

These issues reduce productivity and negatively affect customer satisfaction.

4. Proposed Solution

The proposed solution is the **Shop Easy – E-Commerce Order Management Database System**. The system provides a centralized database that stores all business information securely and efficiently.

The proposed system will:

- Maintain customer records.
- Organize products into categories.
- Track inventory levels.
- Manage suppliers.
- Process customer orders.
- Record payment details.
- Track shipments.
- Store customer reviews.
- Generate business reports.

The system will improve accuracy, reduce redundancy, and support better business decision-making.

5. Project Objectives

The objectives of the proposed system are:

- To maintain accurate customer information.
 - To manage products and categories efficiently.
 - To maintain supplier information.
 - To monitor inventory levels.
 - To simplify order processing.
 - To manage payment transactions.
 - To track shipments.
 - To store customer reviews.
 - To reduce manual work.
 - To improve operational efficiency.
 - To generate business reports for management.
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6. Purpose of Core Entities

Entity	Purpose
Customer	Stores customer information such as Customer ID, Name, Email, Mobile Number, Address, and Registration Date.

Entity	Purpose
Category	Stores category details used to organize products.
Product	Stores product information including price, stock quantity, and category.
Supplier	Stores supplier information for product procurement.
Order	Stores customer order information.
Order Details	Stores products included in each order along with quantity and unit price.
Payment	Stores payment details for customer orders.
Shipment	Stores shipment information and delivery status.
Review	Stores customer ratings and comments for purchased products.

7. Major Business Processes

The system performs the following business processes:

1. Customer Registration
 2. Product Management
 3. Category Management
 4. Inventory Tracking
 5. Order Processing
 6. Payment Management
 7. Shipment Tracking
 8. Customer Review Management
 9. Business Report Generation
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8. Stakeholder Analysis

Stakeholder	Responsibilities
Customer	Registers, places orders, makes payments, tracks shipments, submits reviews.
Administrator	Manages products, customers, orders, suppliers, inventory, and reports.
Supplier	Supplies products to the company.
Warehouse Staff	Updates stock and prepares products for shipment.
Delivery Staff	Delivers customer orders and updates delivery status.
Finance Team	Verifies and records payments.
Management	Reviews reports and monitors business performance.

9. Functional Requirements

The system shall:

- Register customers.
 - Maintain customer records.
 - Manage product categories.
 - Manage products.
 - Manage suppliers.
 - Update inventory.
 - Process customer orders.
 - Record order details.
 - Manage payments.
 - Track shipments.
 - Store customer reviews.
 - Generate business reports.
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10. Non-Functional Requirements

The system should provide:

- Security
 - Reliability
 - Scalability
 - Availability
 - Data Integrity
 - High Performance
 - User-Friendly Interface
 - Maintainability
 - Backup and Recovery
 - Fast Response Time
 - Compatibility
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11. Expected Benefits

The proposed system offers the following benefits:

- Centralized data management.
- Reduced data redundancy.
- Improved inventory management.
- Faster order processing.
- Accurate payment tracking.
- Better shipment monitoring.
- Enhanced customer satisfaction.
- Improved reporting and decision-making.
- Increased operational efficiency.

12. Conclusion

The **Shop Easy – E-Commerce Order Management Database System** provides a reliable and centralized solution for managing all major e-commerce operations. By integrating customer management, product management, inventory tracking, order processing, payment management, shipment tracking, and customer reviews into a single database, the organization can improve efficiency, accuracy, and overall business performance. This requirement analysis serves as the foundation for the subsequent stages of database design, ER modeling, normalization, SQL implementation, and report generation.